SUPPORTING ADULT GUIDANCE

# Recurring decimals as fractions

Number Systems and Calculator Judgement · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

## What this lesson does

This lesson explains why some fractions give recurring decimals and how to convert them back. Learners predict terminating or recurring and discover the prime-factor rule, line up repeating tails to see why subtraction leaves a whole number, watch remainders on a wheel to see why repetition is inevitable, and solve decimal mysteries including $0.\dot{9} = 1$.

Driving Question: *Can a decimal go on for ever and still be a fraction?*

## The mathematics, in plain terms

In long division by d, each remainder lies in {0, 1, …, d – 1}. If a remainder is 0 the decimal terminates; otherwise a remainder must recur within d – 1 steps, and the digits recur from that point. A fraction in lowest terms terminates exactly when its denominator has no prime factors other than 2 and 5. A recurring decimal $x$ with a repeating block of length k satisfies (10^k – 1)$x$ = an integer (after adjusting for any non-repeating part), so $x$ is rational.

## The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

**Example 1 — Why the digits repeat**

Work out 3/7 as a decimal, watching the remainders.

| | |
|---|---|
| **1.** 30 ÷ 7 = 4 r 2, then 20 ÷ 7 = 2 r 6, 60 ÷ 7 = 8 r 4 … | digits 4, 2, 8 … |
| **2.** … 40 ÷ 7 = 5 r 5, 50 ÷ 7 = 7 r 1, 10 ÷ 7 = 1 r 3 | digits 5, 7, 1 |
| **3.** Remainder 3 is where we started, so everything repeats. | 0.428571428571… |

**3/7 = $0.\dot{4}2857\dot{1}$. Only 6 non-zero remainders are possible, so the block has at most 6 digits.**

**Example 2 — Line up the tails**

Write $0.\dot{4}\dot{5}$ (0.4545…) as a fraction.

| | |
|---|---|
| **1.** Two digits repeat, so multiply by 100. | $x$ = 0.4545…, 100$x$ = 45.4545… |
| **2.** Subtract: the tails cancel. | 100$x$ − $x$ = 45, so 99$x$ = 45 |
| **3.** Divide and simplify. | $x$ = 45/99 = 5/11 |
| **4.** Check: 5 ÷ 11 = 0.4545… ✓ | ✓ |

**$0.\dot{4}\dot{5}$ = 5/11.**

## Why before how

The remainder wheel answers 'why does it repeat?' before the method answers 'how do I convert it?'. In Line up the tails, learners choose the multiplier themselves and see when the tails match — so the rule 'multiply by 10 to the power of the block length' is discovered rather than memorised. Calculators are not used: they hide the repeat.

## The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

| Cornerstone | Its role here | Invitational timing |
|---|---|---|
| Connection | Terminate or recur? Predict, check, and discover the 2s-and-5s rule. | ~7 min |
| Movement | Line up the tails: choose the multiplier so the repeating tails cancel. | ~11 min |
| Reflection | The remainder wheel: watch the remainders until one comes back. | ~14 min |
| Creativity | The decimal detective: 1/7, 1/11, $0.\dot{9}$ = 1, and make your own. | ~8 min |
| Rest | Remainders going round like the hands of a clock. | ~2 min |
| Nutrition | Intellectual: certainty about something that goes on for ever. | ~3 min |

## Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

**Judging terminating or recurring without simplifying (9/24 looks like it recurs).**
Simplify first: 9/24 = 3/8.

**Multiplying by 10 when two digits repeat.**
Look at the two lines: do the tails match digit for digit?

**Including a non-repeating digit in the repeating block (0.1666… as 16 repeating).**
Ask which digits come round again, and use two multiples of $x$.

**Believing $0.\dot{9}$ is slightly less than 1.**
Ask what number lies between them; then use the method.

**Rounding a recurring decimal and calling it exact.**
Compare 0.33 with 1/3: multiply each by 3.

## Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

**Please hold this line**
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

## Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: changing fractions to decimals by division, and simplifying fractions. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

## Supporting through questions: an example

Suppose a learner says 5/6 = $0.8\dot{3}8\dot{3}$… (thinking 83 repeats). The aim is a question that lets them find the slip themselves.

- "Divide 5 by 6 slowly. Which remainder comes back?"
- "Does the 8 appear again?"

- “Which digits repeat after the first step?”
- “How would you write that with dots?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

## Questions you might have

**Is $0.\dot{9}$ really equal to 1?**
Yes. There is no number between them, and the algebra gives exactly 1. It is a good example of one number having two decimal names.

**Why no calculators?**
A calculator shows a rounded decimal, which hides the repeat and the remainders that explain it.

**Are unsimplified fractions accepted?**
Where a question asks for the simplest form, learners are nudged to simplify. The value is recognised as right.

**What comes next?**
Lesson 3: order of operations and calculator judgement.

## Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

## Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

| Where they are | What it can look like |
|---|---|
| Approaching Proficiency | Beginning to engage; names some features and makes a start with support. |
| Developing Proficiency | Carries out the method with prompts; can explain part of the reasoning. |
| Proficiency | Works through independently and explains why each step is true. |

| | |
|---|---|
| Refining Proficiency | Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else. |

## A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

## If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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